

Atoms and Molecules:

- Around 500 B.C., Indian Philosopher named Maharishi Kanad postulated that matter can be divided into extremely small particles, which can't be divided further and are called as Parmanus.
- Greek Philosophers - Democritus and Leucippus gave same idea but called these indivisible particles as "Atoms".

Atom - "That can't be cut".

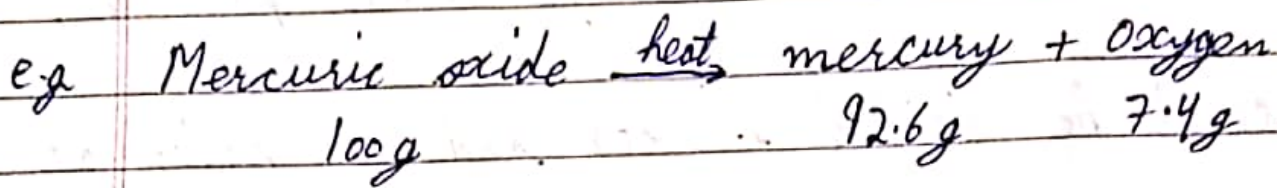
→ Laws of Chemical Combination:

- 1) Law of Conservation of mass: (By Antoine Lavoisier)
- During any physical or chemical change

the total mass of reactants is equal to total mass of products.

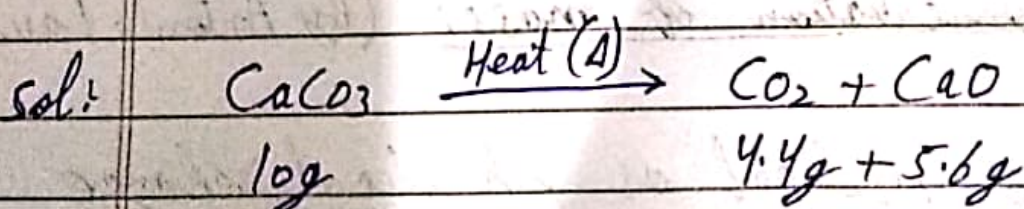
'OR'

"Mass can neither be created nor destroyed".



$$\begin{aligned} \text{Total mass of Reactants} &= \text{Total Mass of Products} \\ 100\text{g} &= 92.6 + 7.4 = 100\text{g} \end{aligned}$$

Q) If 10g of calcium carbonate (CaCO_3) on heating gave 4.4g of carbon dioxide (CO_2) and 5.6g of calcium oxide (CaO). Show these observations are in agreement with law of conservation of mass?



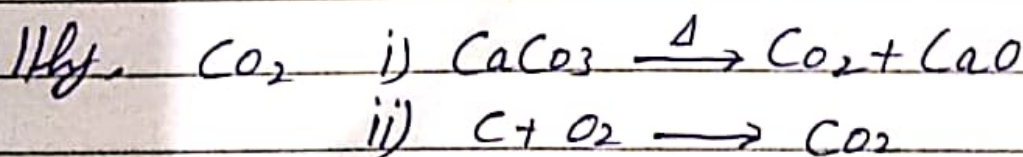
- Acc. to Law of conservation of mass of reactants = Total mass of products.

$$10g = 4.4 + 5.6$$

- Yes it follows Law of Conservation of mass.
- Law of Constant Composition 'OR' Definite Proportion
- ★ - Proposed by Joseph Proust.
- A pure chemical compound is always made up of same elements that are combined in same proportion by mass.

e.g. Water from any source - River, Lake, Ocean, Rain, etc. is always made up of Hydrogen and Oxygen in ratio H_2O

$$2:16 = 1:8$$



In case (i) and (ii) CO_2 prepared by different method but elements are same and ratio is same.

